https://github.com/Et1enneee/comp3500-logistics

# COMP 3500SEF · Final Report

**Warehouse Logistics Management System**
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## 1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The local application serves one warehouse using Next.js, TypeScript, Prisma and SQLite. Real GPS, barcode scanning, recipient telephone collection, payment and carrier integration are outside scope. Deployment is local-only; data integrity is enforced by transactional guards (§5). Requirements and models link to evidence; §10 collects the trail.

## 2. Development process

Plan-driven increments have triggers and verified outcomes; integration and continuous integration provide feedback. Product or specification changes form increments; evidence production belongs to hardening.

**Table A. Increment, trigger and outcome**

| Increment and date | Trigger | Outcome |
|---|---|---|
| Planning, 9 and 11 Oct 2026 | Project selection | Responsibilities and contracts recorded |
| Test-led review, 17 Oct 2026 | Assess order and stock functions | Defects recorded; drove confirmation |
| Confirmation, 26–28 Oct 2026 | Failed endpoint and missing ledger | Transaction verified |
| Integrated application, 1 Nov 2026 | Integrate foundation | Customer, staff and admin workflows |
| Driver and privacy, 2 Nov 2026 | Delivery and data minimisation | Assigned tasks and privacy checks |
| Documentation hardening, 10–11 Nov 2026 | Post-presentation model and evidence review | Diagrams and traceability aligned |
| Export simplification, 11 Nov 2026 | Scripted walkthrough finding (issue #15) | Selected-date export in three actions; regression passed |

The workflow, issue tracker and increment register record integration, problems and context.

## 3. Requirements

The specification defines obligations and rationale.

**Functional requirements and verification**

| FR | The system shall | Rationale | Verification |
|---|---|---|---|
| FR-01 | create orders with product lines and totals | start fulfilment | Test |
| FR-02 | enforce lifecycle and stock reservation or release | protect invariants | Test |
| FR-03 | number orders and shipments and show owner tracking | retrieve progress | Test |
| FR-04 | show inventory, replenishment, ledger and low-stock alerts | support replenishment | Test |
| FR-05 | record event notifications and allow reading | communicate progress | Test |
| FR-06 | provide admin aggregates and CSV export | support review | Test |
| FR-07 | authenticate and enforce role and object access | protect data | Test |
| FR-08 | assign shipped orders and simulate delivery | complete handoff | Test |

Non-functional requirements cover NFR-01 performance; NFR-02 integration-point availability; NFR-03 least privilege; NFR-04 scripted operation effort; NFR-05 process quality. Their criteria and measurement are defined in the specification.

**Table B. Risk coverage** (detailed mapping)

| Risk | Addressed by | Treatment status |
|---|---|---|
| RK-01 scheduling slip | Increment register | Managed — scheduling plan |
| RK-02 contract rework | FR-02, NFR-05 | Mitigated — transition tests |
| RK-03 test-week capacity | Priority register | Managed — scope priorities |
| RK-04 setup delay | NFR-05 | Managed — CI workflow |
| RK-05 demo data | FR-01/04/06, NFR-05 | Managed — seed regression test |
| RK-06 assignments and test squeeze | Risk plan | Managed — assessment schedule |
| RK-07 stack unfamiliarity | NFR-05 support | Managed — setup guide |
| RK-08 report and video compete for time | Risk plan | Managed — report and video plan |

Mitigated records a mitigation; managed records a mechanism while the historical register remains open.

## 4. System models

Five views link roles, lifecycle, data, transaction, layers. Local transactions preserve consistency without splitting services; state and stock guards enforce rules.

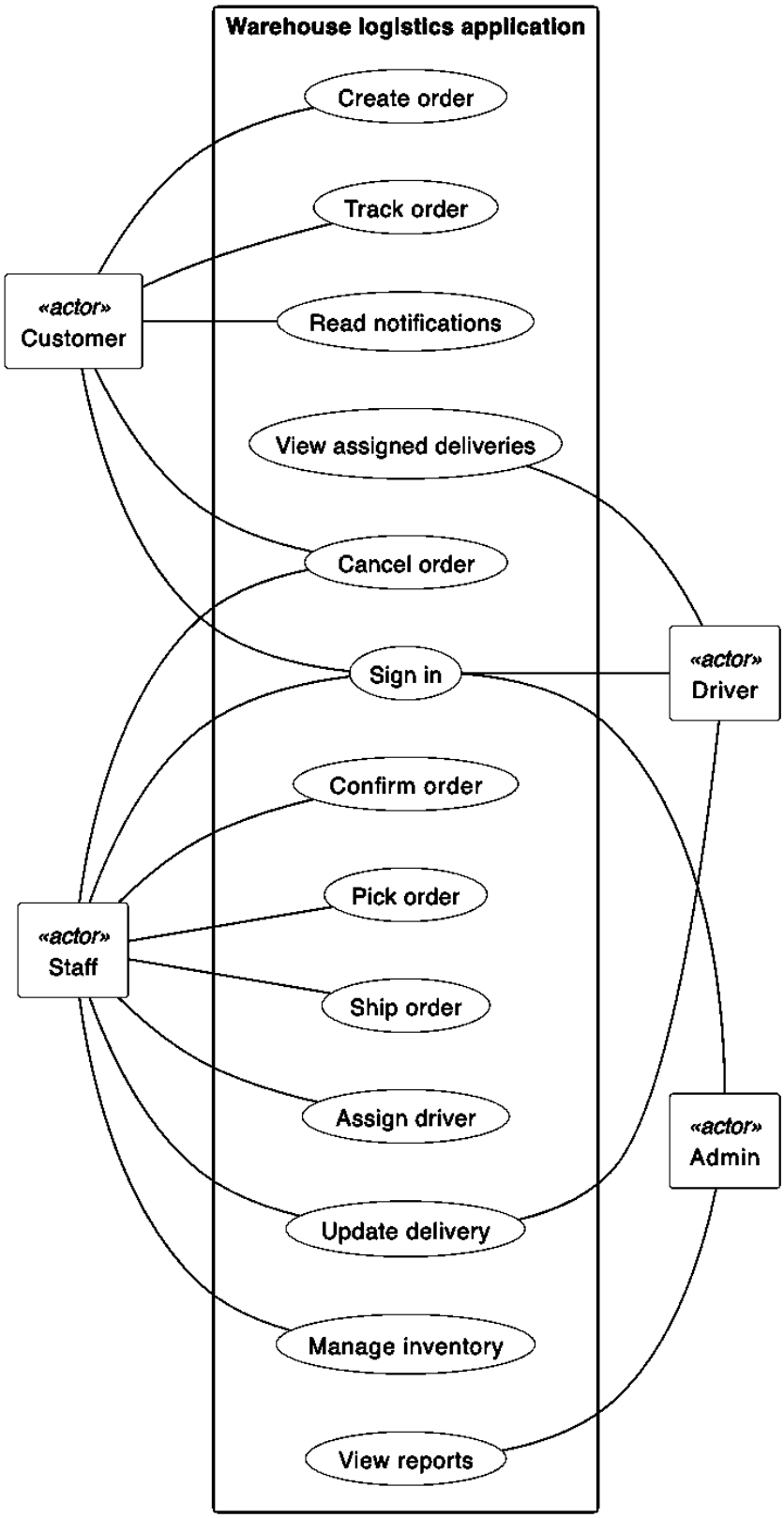



*Figure 1. Which roles perform each function? Rebuilt from the reviewed draft; §9.* Source.

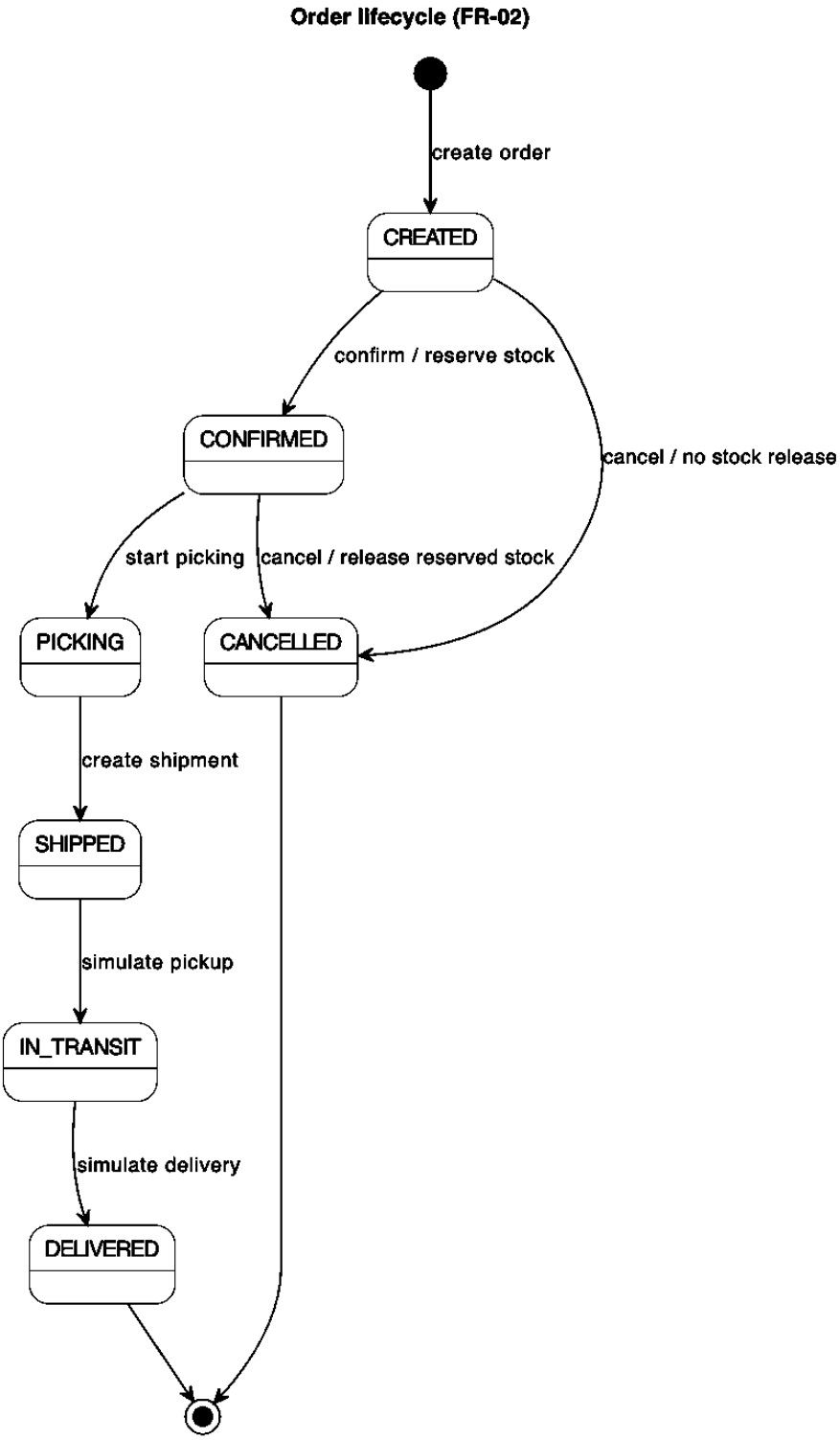



*Figure 2. Which order transitions and stock effects are valid?* Source.

Each item references one product; each order has 0..1 shipment; each product has 0..* inventory logs.

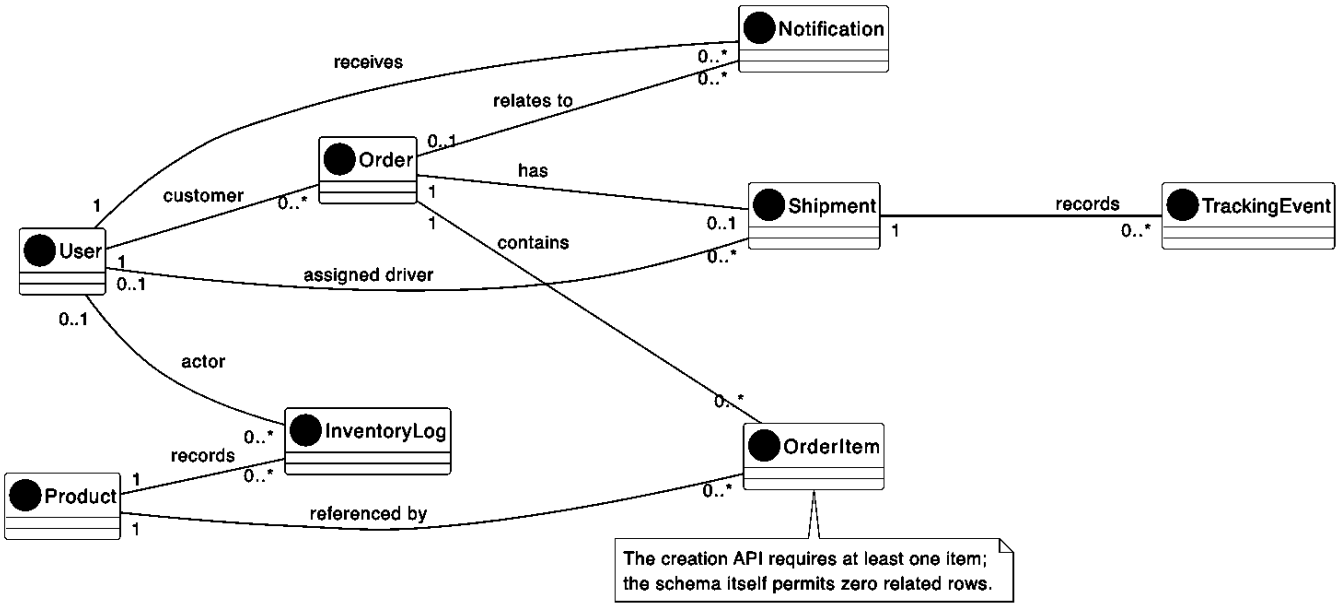



*Figure 3. Data relationships: the schema permits zero items; the creation API requires at least one.* Source.

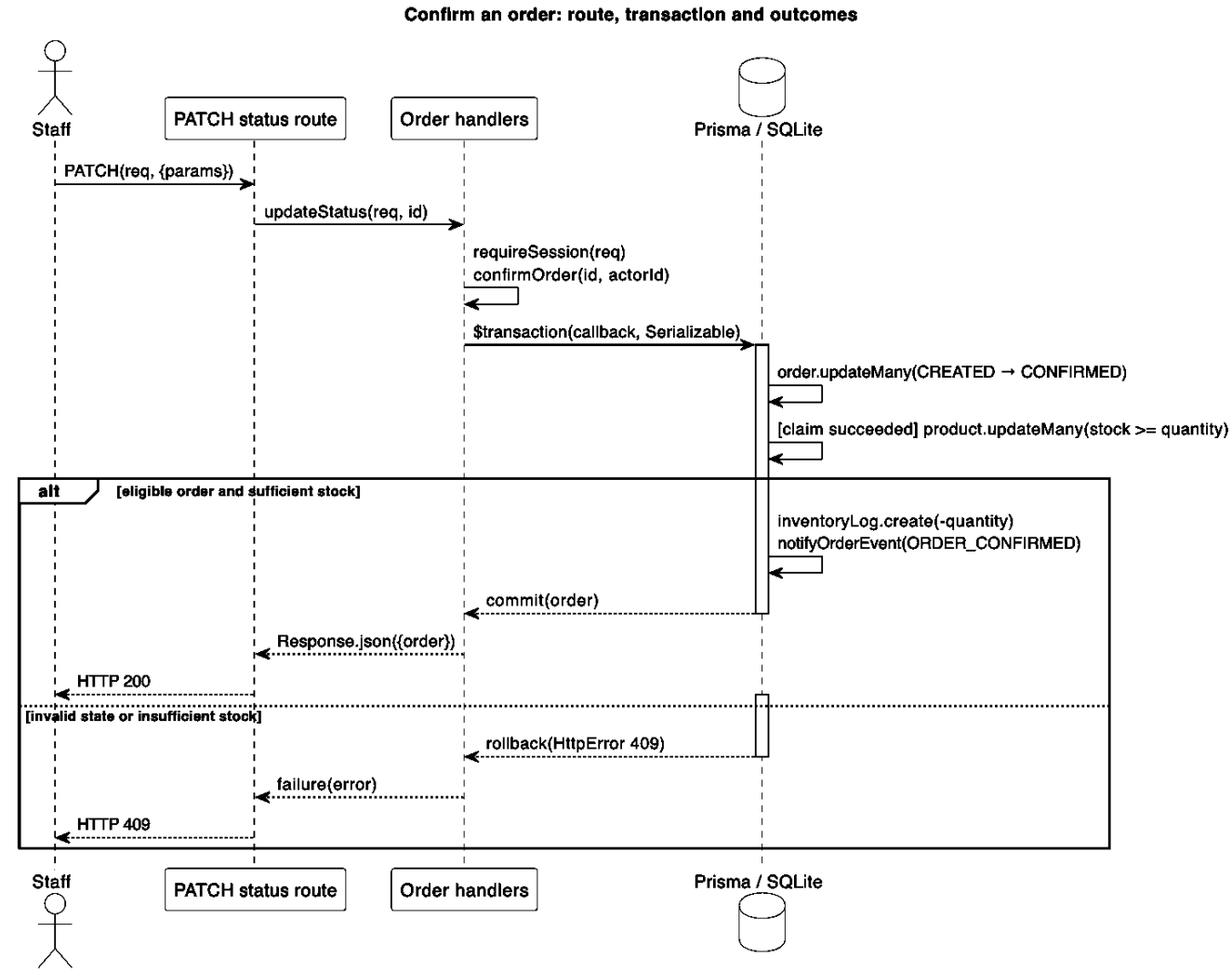



*Figure 4. How does confirmation commit or roll back?* Source.

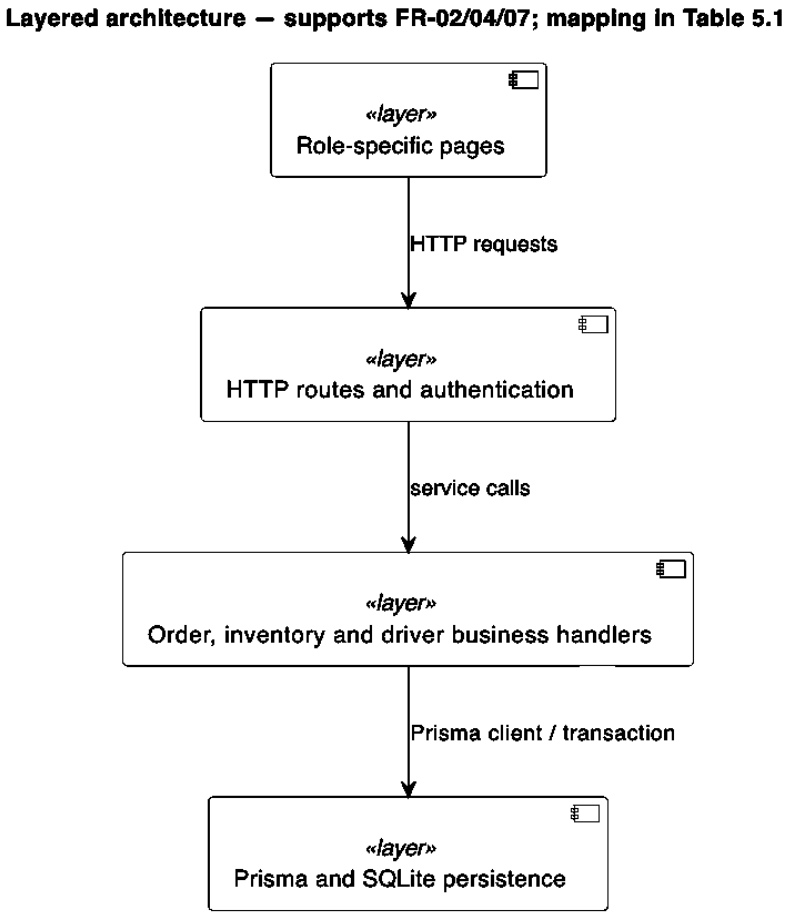



*Figure 5. How do four layers connect?* Source.

## 5. Implementation

Design rules map to implementation and tests; staff cancellation requires CONFIRMED.

| Design | Implementation | Verification |
|---|---|---|
| Guards: CREATED and sufficient stock | order-api.ts:250–253, 276–280 | Orders |
| Serializable state, stock, ledger and notification writes | order-api.ts:245–293 | Orders |
| driverId: assign only SHIPPED; retained after delivery | driver-api.ts:44–51; transition | Driver |
| Selected-date export | admin/page.tsx:24–48 | Browser |

## 6. Verification and validation

Verification checks requirements; validation assesses fitness for users. The regression log and CI show 38 unit, 140 integration, 2 delivery (driver simulation) and 26 browser checks.

| Boundary case | Observed | Test |
|---|---|---|
| Repeat confirmation | HTTP 409; state and stock unchanged | Orders |
| Stock equals demand | Confirmation succeeds; stock becomes zero | Orders |
| Stock 4 versus threshold 5 | Alert; stock 5 is not low | Inventory |

Insufficient stock returns HTTP 409 with unchanged state and stock. In a local-loopback measurement recorded on 11 November 2026, API p95 was 3.2–13.6 ms and first-contentful-paint p95 was 64–96 ms, against NFR-01 targets of at most 2 s and 3 s respectively. NFR-02 records 40 successful build–test–browser runs (observations).

## 7. Evolution

The early defect record drove the confirmation transaction and retest; the post-presentation audit corrected models and renewed tests. A four-action administrator task became three actions including date entry (regression), or one click from a prepared screen (not scripted).

## 8. Validation

The automated walkthrough counts at most three scripted actions across eleven selected operations from prepared screens (excluding login, navigation and recovery). The human protocol defines role tasks, completion, errors, help requests and time, with acceptance criteria defined before each run. No participant sessions were recorded before submission. Validation rests on those eleven scripted operations.

## 9. Feedback and reconciliation

**Table C. Feedback to evidence**

| Feedback | Disposition | Evidence |
|---|---|---|
| Diagrams | Corrected models and top-down architecture | Compliance matrix |
| FR-04 | Stock transition effects belong to FR-02 | Traceability |
| Increments | Triggers and outcomes | Register |
| Presentation p95 | The recorded probe is the authoritative measurement | Measurement |
| Admin task | Export re-tested; no participant session recorded (§8) | Issue #15 |
| Unit count | Two soak-summary tests increased unit count from 36 to 38 | Test history |
| NFR scope | Criteria revised for local deployment | SRS revision log |

Both goals connect order decisions and role-separated workflows to evidence.

## 10. Evidence

Traceability · Specification · Evidence directory · Baseline record.

Full evidence trail: docs/final-report.md in the repository (evidence baseline `6eb01afb1798c7361205b10597d80c0c26e8fdf6`).